Robot description

planar 4-DOF serial robot arm, with 4 revolute joints (R-R-R-R). The robot operates in the XZ plane.

The generalized coordinates are:

$$q_{(t)}=[q_1(t)+q_2(t)+q_3(t)+q_4(t)]^T$$

Link lengths (m):

$$L=[0.35,0.3,0.25,0.2]$$

Masses (kg):

$$m=[2.0,1.6,1.2,0.8]$$

Center of mass :

$$l_{c,i}=\frac{l_i}{2}$$

where li is the link length of link (i)

defining the comulative orientation of link (i) in the XZ plane as:

$$\theta_i=\sum_{k=1}^{i} q_k$$

Let x_ji z_ji be position of joint (i) in the world frame. Then the COM(center of mass) position of link (i) is:

$$x_{c,i}=x_{Ji}+l_{c,i}\cos(\theta_i),\ z_{c,i}=z_{Ji}+l_{c,i}\sin(\theta_i)$$

with the base joint at:

$$(x_{J1},z_{J1})=(0,0)$$

These COM positions are used to compute the gravity potential energy

$$V_{(q)}=\sum_{i=1}^{4} m_i g_0 z_{c,i}(q)$$

and the COM liniar-velocity jacobians needed to assemble the inertia matrix Mq.

Link inertias (slender rods about COM)

$$I_i = \frac{m_i L_i^2}{12}$$

Rviz video attached of the provided robot.

Dynamical Equations of Motion

The arm using standart rigid-body manipulator dynamics in the vertical plane:

$$M(q)\ddot{q}+c(q,\dot{q})+g(q)+D\dot{q}=\tau$$

Where:

q joint angles (rad)
$\dot{q}$ joint velocities (rad/s)
$\ddot{q}$ joint accelerations (rad/s\(^2\))
τ joint torque inputs (Nm)
$M(q)$ inertia matrix (4×4)
$c(q,\dot{q})$ Coriolis/centrifugal vector (4×1) (disabled in the project to simplify the model and improve numerical robustness)
$g(q)$ gravity vector (4×1)
D viscous damping matrix (4×4)

The accelerations are computed by solving:

$$\ddot{q}=M(q)^{-1}(\tau-c(q,\dot{q})-g(q)-D\dot{q})$$

The inertia matrix is assembled using Jacobians of each link COM in the XZ plane :

$$M(q)=\sum_{i=1}^{4}(m_i J_{vi}(q)^T J_{vi}(q)+I_i J_{wi}^T J_{wi})$$

where:
J_vi is the 2x4 linear velocity Jacobian of COM in XZ and J_wi is the 1x4 angular velocity jacobian for rotation around z.

Gravity term g(q):

potential energy is:

$$V_{(q)}=\sum_{i=1}^{4} m_i g_0 z_{c,i}(q)$$

where:
z_ci(q) is the COM height of link (i) and g_0 = -9.81 (m/s^2)

the gravity vector:

$$g(q)=\frac{\partial V(q)}{\partial q}$$

in implementation g(q) computed numerically using central differences.

<u>Damping term:</u>

Viscous joint damping modeled as:

$$D\dot{q}, D=diag(4.0, 3.0, 2.0, 1.6)$$

<u>Coriolis/centrifugal term:</u>
For simplicity and robustness, the term was optionally disabled in the simulation.

These equations represent 4 coupled, time dependent differential equations (second order).

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Numerical simulation/integration

The system simulated by integrating the state:

$$x(t)=\begin{bmatrix} q(t) \\ \dot{q}(t) \end{bmatrix}$$

with dynamics:

$$\dot{x}(t)=\begin{bmatrix} \dot{q}(t) \\ \ddot{q}(t) \end{bmatrix}$$

and

$$\ddot{q}=M(q)^{-1}(\tau-c(q,\dot{q})-g(q)-D\dot{q})$$

Numerocal integration was performed using 4th order Runge Kutta with timestep deltat=0.002 s in other words operating in 500Hz

Initial condition (start position with zero velocity)

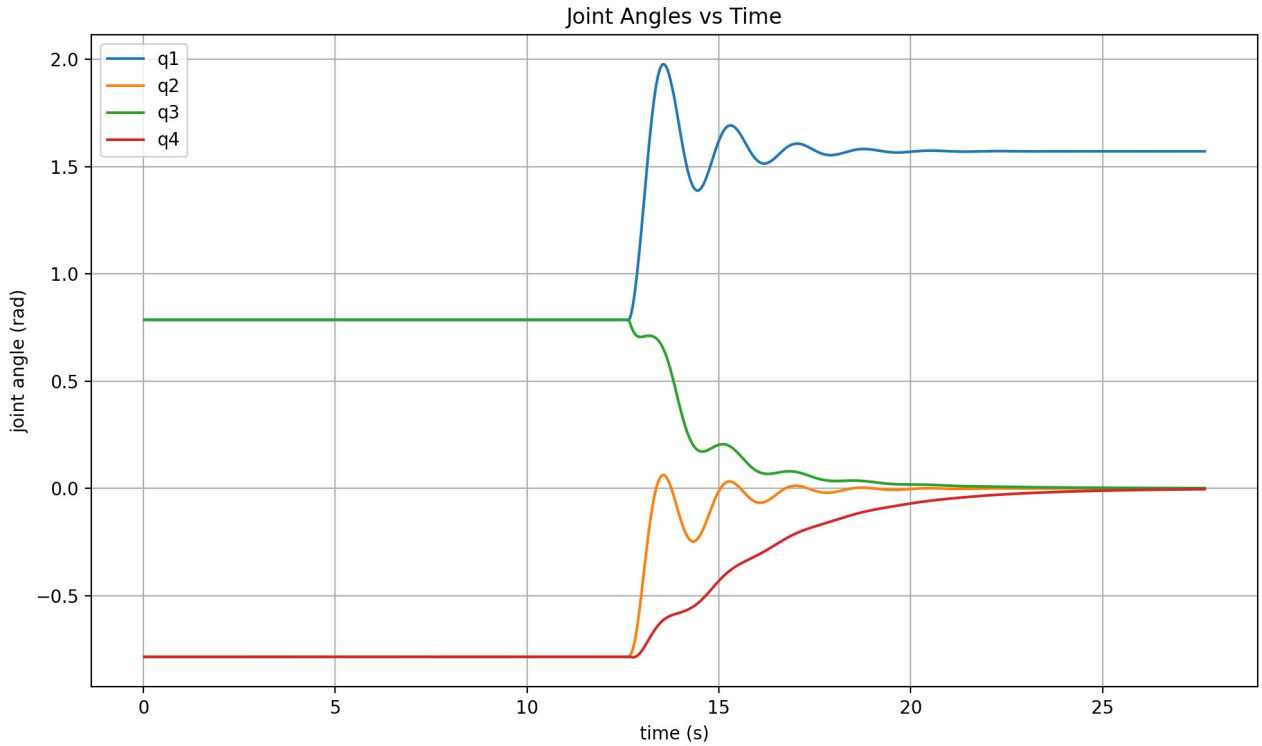
The simulation starts from:

$$q(0)=[\frac{\pi}{4},-\frac{\pi}{4},\frac{\pi}{4},-\frac{\pi}{4}]^T rad$$

$$\dot{q}(0)=[0,0,0,0,]^T rad/s$$

Zero-input experiment ($\tau=0$ motion graphs+video in description)

Simulated the arm with zero input torques, starting from the above initial conditions, the arm exhibits gravity-driven motion. Including transient oscillation, and settles to a static equillibrium because of damping. The q_i(t) (joints angles in radians) were logged and plotted vs time. And video is provided for visualization.



Initial phase:
Joints start at their given initial angles (`q1 ≈ 0.8`, `q2 ≈ -0.8`, `q3 ≈ 0.8`, `q4 ≈ -0.8` radians = ±45°).
Transition:
Around 12–14 seconds, the system is allowed to move freely
The joints begin to swing and oscillate under gravity, showing dynamic response and then damping out.

Target position selection

End-effector target was commanded:

$$[x, z, \theta]_{des} = [0.6, -0.2, 0.1]$$

where x,z in meters and theta in radians.

An Inverse kinematics produced the following joint-space target:

$$q_{des} = [0.06824942 \; -2.35802943 \; 2.2175367 \; 0.17222482] \, rad$$

from that we can see that sum of all thetas gives ~0.1 rad close to the target theta.
Moreover from FK we get x~0.6 m and z~-0.2 m from the q_des.

Pid controller design

A joint-space PID controller with angle wrapping to avoid discontinuities.
The control torque is:

$$\tau = \tau_{PID} + g(q)$$

$$\tau_{PID} = K_p e + K_d \dot{e} + K_i \int e \, dt$$

where e=q_des-q (wrapped to -pi to pi), and dot e = -dot q
The gains used were diagonal matrices:

$$K_p = diag[30,25,18,10]$$
$$K_d = diag[18,15,12,9]$$
$$K_i = diag[0,0,0,0]$$

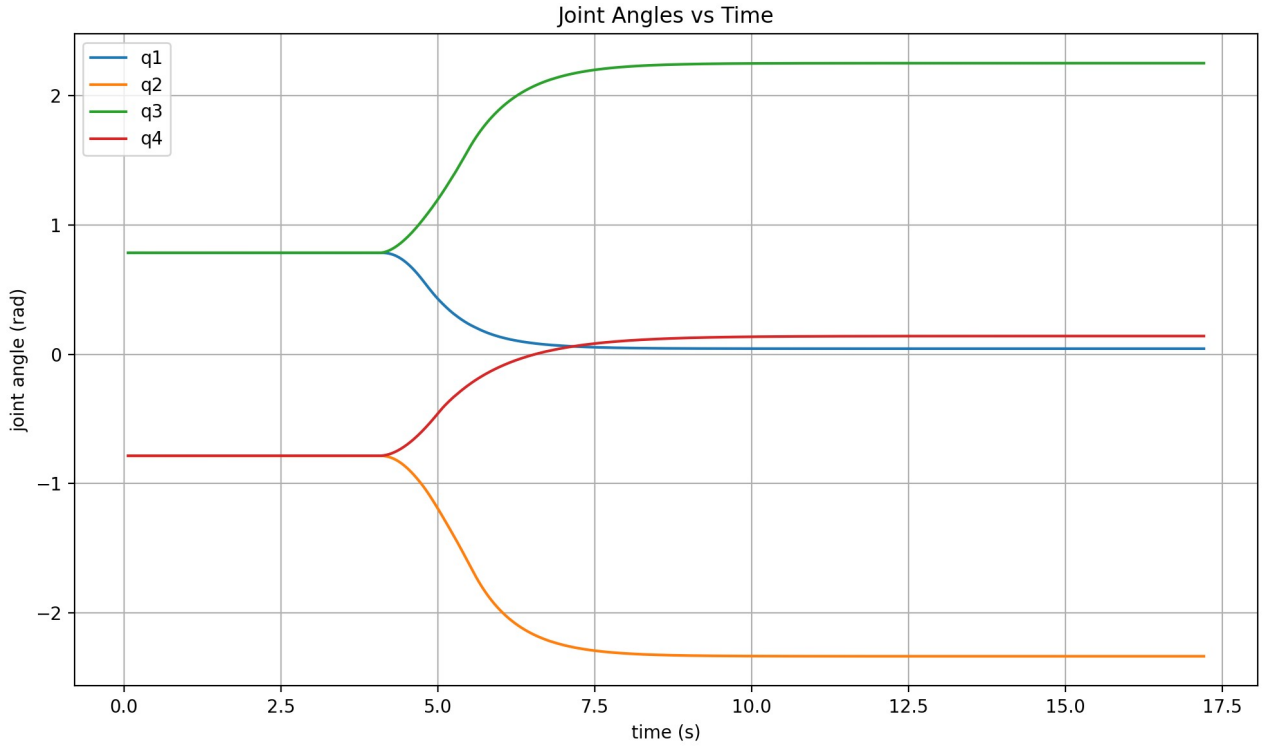
I didn’t use the integral term (Ki) in my controller because it caused instability and oscillations. Since my robot’s dynamics are well-modeled and the PD+gravity compensation is accurate, the term was unnecessary and actually degraded performance. I keep the Ki for future coriolis that i kept in the code but couldnt implement with good results.

Additional implementation details:
torque saturation limits(-tau_max to +tau_max) with tau_max=[20,15,10,7] Nm

target rate-limiting was used to avoid commanding abrupt joint steps.

Pid experiment: motion graphs + video in description

after commanding the target, the closed-loop system transitions from holding initial pose to moving toward the desired configuration. The joint angles converge to target values with stable behavior and small steady-state error.
Command to go to (x,z,theta)=(6,-2,0.1)



Initial phase:
Joints start at their given initial angles (`q1 ≈ 0.8`, `q2 ≈ -0.8`, `q3 ≈ 0.8`, `q4 ≈ -0.8` radians = ±45°).
Transition:
Around 3–4 seconds, the system is start to move to the target
The joints showing dynamic response antill it get to the target position.
New IK target received (rad): [0.06824942 -2.35802943 2.2175367 0.17222482]
FK:

$$L=[0.35,0.3,0.25,0.2]$$
$$\theta=\theta_1+\theta_2+\theta_3+\theta_4=0.105_{rad}$$
$$x=l_1*\cos(\theta_1)+l_2*\cos(\theta_1+\theta_2)+l_3*\cos(\theta_1+\theta_2+\theta_3)+l_4*\cos(\theta_1+\theta_2+\theta_3+\theta_4)\approx 0.6$$
$$z=l_1*\sin(\theta_1)+l_2*\sin(\theta_1+\theta_2)+l_3*\sin(\theta_1+\theta_2+\theta_3)+l_4*\sin(\theta_1+\theta_2+\theta_3+\theta_4)\approx -0.2$$

another check from the bash:


```
layout:
  dim: []
  data_offset: 0
data:
- 0.6000303043223334
- 0.19997259436140266
- 0.10004494026009736
```